

Linear Optimization (CS5040)

Assignment Questions

Assignment 1

Assumptions

1. Polytope is non-degenerate.
2. Polytope is bounded.
3. Rank of A is n .
4. Initial feasible point is given.

Implement the simplex algorithm to maximize the objective function. You need to implement the method discussed in class.

Input: CSV file with $m + 2$ rows and $n + 1$ columns.

- The first row excluding the last element is the initial feasible point z of length n .
- The second row excluding the last element is the cost vector c of length n .
- The last column excluding the top two elements is the constraint vector b of length m .
- Rows third to $m + 2$ and columns one to n form the matrix A of size $m \times n$.

Output: You need to print the sequence of vertices visited and the value of the objective function at that vertex.

Instructions:

1. You need to implement in Google Colab and share the link.
2. Implementation language is Python.
3. Input CSV file name is `Testcase.csv` from the Colab workspace.

Assignment 2

Assumptions

1. Initial feasible point is given.

Implement the simplex algorithm to maximize the objective function. You need to implement the method discussed in class.

Input: CSV file with $m + 2$ rows and $n + 1$ columns.

- The first row excluding the last element is the initial feasible point z of length n .
- The second row excluding the last element is the cost vector c of length n .
- The last column excluding the top two elements is the constraint vector b of length m .
- Rows third to $m + 2$ and columns one to n form the matrix A of size $m \times n$.

Output: You need to print the sequence of vertices visited and the value of the objective function at that vertex.

Instructions:

1. You need to implement in Google Colab and share the link.
2. Implementation language is Python.
3. Input CSV file name is `Testcase.csv` from the Colab workspace.

Assignment 3

Implement the simplex algorithm to maximize the objective function. You need to implement the method discussed in class.

Input: CSV file with $m + 1$ rows and $n + 1$ columns.

- The first row excluding the last element is the cost vector c of length n .
- The last column excluding the top element is the constraint vector b of length m .
- Second to $m + 1$ rows and columns one to n form the matrix A of size $m \times n$.

Output: You need to print the initial feasible point, sequence of vertices visited, and the value of the objective function at that vertex.

Instructions:

1. You need to implement in Google Colab and share the link.
2. Implementation language is Python.
3. Input CSV file name is `Testcase.csv` from the Colab workspace.

Assignment 4

Implement an algorithm to find the minimum weight perfect matching in an edge weighted bipartite graph. You need to implement the method discussed in class.

Input: CSV file with 3 rows and number of columns equal to the number of edges.

- Each column corresponds to an edge.
- First and second rows are the vertex ids of the edges, and the third row is the edge weights.
- Note that vertex ids are integers.

Output: You need to print the values of the dual variables at each iteration.

Instructions:

1. You need to implement in Google Colab and share the link.
2. Max two members in a group. You need to mention the roll numbers of the group members at the top of the code.

Assignment 5

Implement an algorithm to find the minimum cost max flow in a network. You need to implement the method discussed in class.

Input: CSV file with 4 rows and number of columns equal to the number of edges.

- Each column corresponds to an edge.
- First row is the 'from' vertex and second row is the 'to' vertex of the edge, third row is the capacity of the edge, and fourth row is the cost of sending one unit of flow on it.
- Note that vertex ids are integers.
- Source vertex id is '1' and destination vertex id is '2'.

Output: You need to print the total cost at each iteration.

Instructions:

1. You need to implement in Google Colab and share the link.
2. Max two members in a group. You need to mention the roll numbers of the group members at the top of the code.